



P101 · Unit 1 reference

Physical quantities and executable models

Version 1.0 · 3 October 2026 · Quantities, units and the model

Recall fluently · Understand the reasoning · Consult when needed.

Use	Idea and relationship	Conditions / reminders
Recall	Physical value Quantity + numerical value + unit; add reference choices and procedure when they affect meaning.	A position needs an origin and positive direction; a clock reading is not automatically a duration.
Recall	Common unit factors $1 \text{ km} = 1000 \text{ m}$; $1 \text{ m} = 100 \text{ cm} = 1000 \text{ mm}$; $1 \text{ s} = 1000 \text{ ms}$; $1 \text{ min} = 60 \text{ s}$; $1 \text{ kg} = 1000 \text{ g}$.	Convert before comparing. The quantity stays the same while its numerical value changes.
Understand	Areas and ratios Square a length conversion factor for area; cube it for volume. Ratios of like quantities become dimensionless in common units.	$1 \text{ mm}^2 = 0.000001 \text{ m}^2$. Do not convert area as though it were length.
Recall	Changes $\Delta t = t - t_0$; $\Delta x = x - x_0$	Initial position and time describe the same event. Negative coordinates/readings can give positive changes.
Understand	Constant-rate model $x = x_0 + v (t - t_0)$	Fixed origin/direction and constant signed rate over the interval. Our forward-only cases require $t \geq t_0$; the formula alone does not enforce that domain.
Understand	Discriminating checks At $t=t_0$: $x=x_0$. With $v=0$: $x=x_0$. Include ordinary, negative-rate and nonzero-origin cases.	Use independent expected answers. Correct units and passing examples do not prove physical validity.



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Physical quantities and executable models

Version 1.0 · 3 October 2026 · Computation and interpretation

Recall fluently · Understand the reasoning · Consult when needed.

Use	Idea and relationship	Conditions / reminders
Consult	Assignment and execution a = expression evaluates now and binds its result. Recalculate dependent results after changing inputs.	Editing a cell does not execute it. Restart, run from the top, inspect and save.
Consult	Decisions and lists Comparisons: ==, !=, <, <=, >, >=. if/else chooses a path. Lists preserve order and repeats; indices start at 0.	A boundary test checks only its stated condition. Empty lists have no first item.
Consult	Loops and retention for item in values: visits successive values. Initialise result lists before the loop; append paired values together.	A repeated request remains repeated. Do not reset a result list inside each iteration.
Consult	Functions def name(parameters): defines an indented body. return value sends a result to the caller and ends the call.	print displays; it is not a substitute for return. Falling off the end returns None, not numerical zero.
Understand	Local working and inputs Pass physical dependencies explicitly. Local numeric assignments do not reassign outside names.	Mutable input lists can still be changed. A docstring describes units and responsibilities; it does not enforce them.
Understand	Interpretation Separate stipulated inputs, predictions and observations; report conventions, checks and limits.	A clean run checks reproducibility of execution. Evidence about real motion requires an observation procedure.